

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460877.8302 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.04 +/- 0.042
Transit depth (Rp/Rs)^2	0.16 +/- 0.34 [%]
Orbital Inclination (inc)	89.86 +/- 1.8
Airmass coefficient 1 (a1)	0.934 +/- 0.05
Airmass coefficient 2 (a2)	0.038 +/- 0.081
Scatter in the residuals of the lightcurve fit is	5.28 %
Best Comparison Star	#1 - [606, 284]
Optimal Aperture	2.84
Optimal Annulus	9.9
Transit Duration (day)	0.087 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.04 ± 0.042 vs 0.14762 (deviation 1.25σ)
Duration fit vs published	0.087 d vs 0.1033 d (deviation 0.67σ)
Mid-time O–C	4.1 min
Depth SNR	0.5
Residual scatter	5.28 %

Light curve

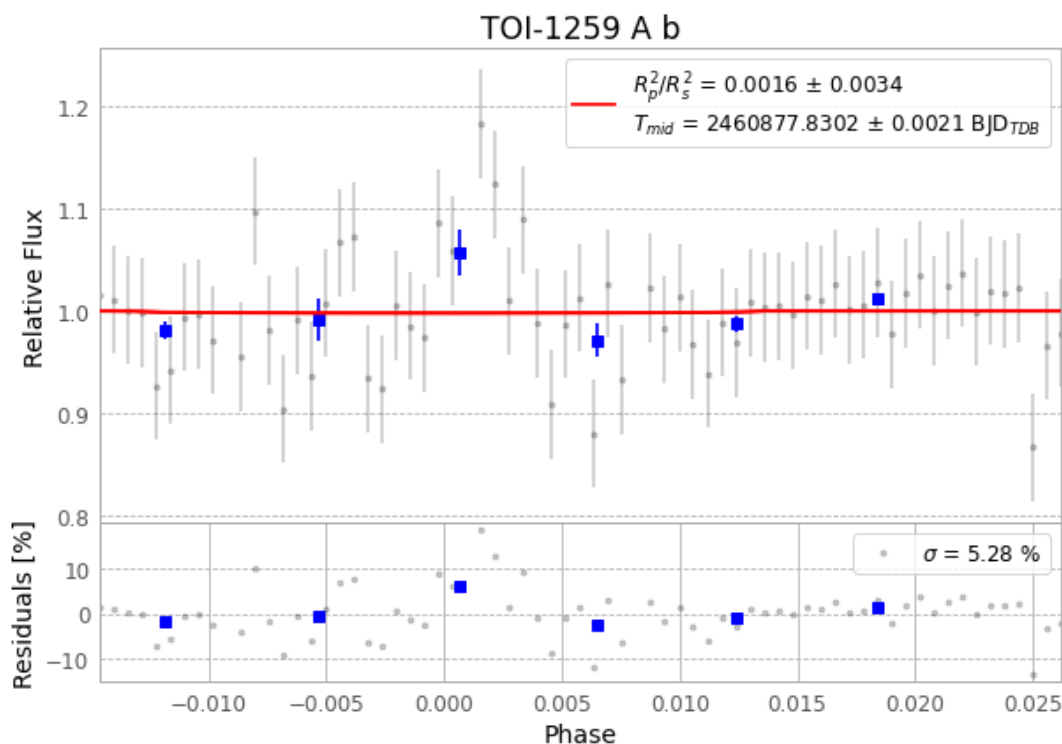


Figure 1 — FinalLightCurve_TOI-1259 A b_2025-07-20.png (SHA-256 6acd114fac61ec78...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 285], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f67724aca5ee5659...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), TOI1259250721064216.FITS → TOI1259250721100616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1259-250721.log (c34bb1f0bf76...), FinalLightCurve_TOI-1259 A b_2025-07-20.png (6acd114fac61...), FinalLightCurve_TOI-1259 A b_2025-07-20.csv (eac16a405965...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:08Z.